

PIMID: A Full-System Simulator for Processing-in-Memory with Intricacy and Diversity

Yuan He, Masaaki Kondo
RIKEN Center for Computational Science
Kobe, Hyogo, Japan
{yuan.he.uw,masaaki.kondo}@riken.jp

Galen M. Shipman, Jered B. Dominguez-Trujillo
Los Alamos National Laboratory
Los Alamos, New Mexico, USA
{gshipman,jerreddt}@lanl.gov

1 INTRODUCTION

As computational demands continue to grow, conventional memory systems struggle to keep pace, prompting the need for innovative approaches that reduce data movement and improve energy efficiency. Recently, Processing-in-Memory (PIM) starts to look promising for this longstanding memory wall challenge and it offers a way forward by integrating computational capabilities directly into memory systems, but current tools for exploring this paradigm remain limited in their scope and flexibility.

To address these shortcomings, we introduce PIMID, a full-system simulation framework tailored for the comprehensive evaluation of PIM architectures. PIMID stands out with its ability to co-simulate host and memory devices, alongside its support for diverse memory technologies. It also allows for detailed and fine-grained configurations of the processing elements (PEs) at various levels plus advanced support for in-memory networks, which enable precise architectural explorations with complex data communication patterns inside the memory.

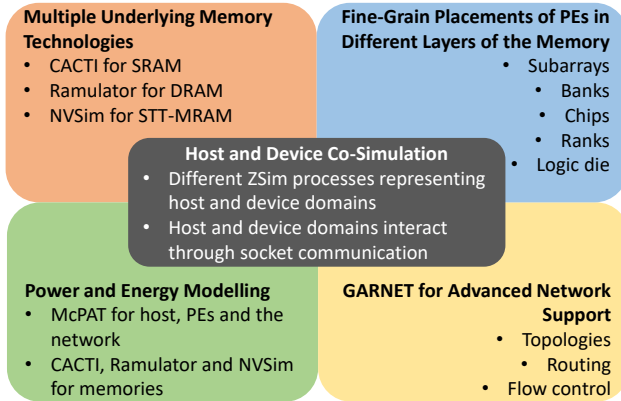


Figure 1: Schematic design and supported features of PIMID.

2 PIMID: DESIGN AND FEATURES

To realize PIMID’s key innovations, we have extended ZSim+Ramulator and MultiPIM as follows (summarized in Fig. 1):

- **Co-Simulation of host and PIM devices:** Distinct ZSim [4] processes are used to represent either host or PIM devices. Interactions between them are implemented with socket-based communication.

- **Multiple memory technologies:** In addition to Ramulator [5], we have integrated CACTI [7] and NVSim [8] to extend the support from DRAM-only to SRAM/STT-MRAM.
- **Fine-grained placements of PEs:** PEs are now allowed to be placed at different layers of the memory, including subarrays, banks, chips, ranks, or the logic die of HBM/HMC. When PEs are placed at subarrays, banks or chips, discrete addressing from the host processor is supported.
- **Power and energy modeling:** Power models from McPAT [9] are employed for the host processor, PEs and the network while Ramulator, CACTI and NVSim are used to estimate power for various memory technologies.
- **Advanced network support in the memory:** Network features, such as topologies, routing and flow control is modeled inside the memory through GARNET 2.0 [10].

3 COMPARISONS BETWEEN MAINSTREAM PIM RESEARCH TOOLS AND PIMID

As shown in Table 1, PIMID is more flexible/comprehensive versus its counterparts. By offering extensive configurability and support for emerging technologies, PIMID serves as a robust foundation for exploring the next generation of memory-centric computing systems. As a next step, we are going to open source it at first; and we also plan to add hardware validation support in the future.

REFERENCES

- [1] UPMEM. *UPMEM SDK*. Available at <https://sdk.upmem.com>.
- [2] B. Hyun et al. Pathfinding Future PIM Architectures by Demystifying a Commercial PIM Technology. In *Proc. of HPCA’24*, pages 263–279, March 2024.
- [3] S. Mosanu et al. PiMulator: a Fast and Flexible Processing-in-Memory Emulation Platform. In *Proc. of DATE’22*, pages 1473–1478, March 2022.
- [4] G. Singh et al. NAPEL: Near-Memory Computing Application Performance Prediction via Ensemble Learning. In *Proc. of 56th DAC*, pages 1–6, June 2019.
- [5] Y. Kim et al. Ramulator: A Fast and Extensible DRAM Simulator. *IEEE CAL*, 15(1):45–49, 2016.
- [6] C. Yu et al. MultiPIM: A Detailed and Configurable Multi-Stack Processing-In-Memory Simulator. *IEEE CAL*, 20(1):54–57, 2021.
- [7] N. Muralimanohar et al. Optimizing NUCA Organizations and Wiring Alternatives for Large Caches with CACTI 6.0. In *Proc. of 40th MICRO*, pages 3–14, December 2007.
- [8] X. Dong et al. NVSim: A Circuit-Level Performance, Energy, and Area Model for Emerging Nonvolatile Memory. *IEEE TCAD*, 31(7):994–1007, 2012.
- [9] S. Li et al. McPAT: An Integrated Power, Area, and Timing Modeling Framework for Multicore and Manycore Architectures. In *Proc. of 42nd MICRO*, pages 469–480, December 2009.
- [10] T. Krishna. *Garnet2.0: A Detailed On-Chip Network Model Inside a Full-System Simulator*. Available at https://bpb-us-e1.wpmucdn.com/sites.gatech.edu/dist/c/332/files/2017/09/Garnet2.0-Tutorial_gem5-workshop_ARM-Rsh-Summit2017.pdf.

	UPMEM SDK [1]	uPIMulator [2]	PiMulator [3]	ZSim+Ramulator [4, 5]	MultiPIM [6]	PIMID (proposed)
Simulation or Emulation Model	Functional	Cycle-accurate	FPGA prototyping	Trace-based	Trace-based	Trace-based
Memory Technology Support	NA	DRAM	DRAM	DRAM	DRAM	DRAM, SRAM, STT-MRAM
Scope of Simulation and Emulation	PIM device	PIM device	Full-system	Host and PIM device, but no co-simulation	Host and PIM device, but no co-simulation	Full-system
PE Placements	Chips	Banks	Banks, chips and ranks	Ranks and logic die	Ranks and logic die	Subarrays, banks, chips, ranks and logic die
Addressing	Discrete	Discrete	Discrete and unified	Unified	Unified	Discrete and unified
VM Support	NA	NA	NA or DIY	NA	PTW	PTW
Network Model	NA	NA	NA or DIY	NA	BookSim	GARNET
Power Model	NA	NA	Measurement	Memory only	Memory only	Host, PE and memory
Hardware Validation	UPMEM RDIMM	UPMEM RDIMM	FPGA	NA	NA	To come

Table 1: Comparisons of Mainstream PIM Research Tools